$\hat{a}^+$ 有没有本征值方程？

$$\hat{a}^+ |a\rangle \overset{?}{=} a |a\rangle .$$

左边：
$$\hat{a}^+ |a\rangle = \sum_{n=0}^{\infty} \hat{a}^+ |n\rangle\langle n| a\rangle$$

利用 $\hat{a}^+ |n\rangle = \sqrt{n+1}\,|n+1\rangle$ ，

$$= \sum_{n=0}^{\infty} \sqrt{n+1}\,|n+1\rangle\langle n|a\rangle$$

$$= |1\rangle\langle 0|a\rangle + \sqrt{2}\,|2\rangle\langle 1|a\rangle + \sqrt{3}\,|3\rangle\langle 2|a\rangle + \cdots\cdots$$

右边：
$$a|a\rangle = a\sum_{n=0}^{\infty} |n\rangle\langle n|a\rangle$$

$$= a|0\rangle\langle 0|a\rangle + a|1\rangle\langle 1|a\rangle + a|2\rangle\langle 2|a\rangle + \cdots\cdots$$

左边 $\neq$ 右边 ：无解！

§1.6 相干态表象

1. 相干态的正交性和完备性 (基矢: $\{|\alpha\rangle\}$)

(a) 正交性：相干态可归一，但不正交

$$\langle\alpha|\alpha\rangle = 1 \quad , \quad \langle\alpha'|\alpha\rangle \neq 0 \quad (\alpha' \neq \alpha)$$

证明：

$$\langle\alpha'|\alpha\rangle = \left(\sum_{m=0}^{\infty} \frac{(\alpha'^*)^m}{\sqrt{m!}}\langle m| \; e^{-\frac{|\alpha'|^2}{2}}\right)$$

$$\left(\sum_{n=0}^{\infty} \frac{(\alpha)^n}{\sqrt{n!}}|n\rangle \; e^{-\frac{|\alpha|^2}{2}}\right)$$

$$= \sum_{n=0}^{\infty} \frac{(\alpha'^*\alpha)^n}{n!} e^{-\frac{|\alpha'|^2}{2}} e^{-\frac{|\alpha|^2}{2}}$$

$$= \exp\left[-\frac{1}{2}\left(|\alpha|^2 + |\alpha'|^2 - 2\alpha'^*\alpha\right)\right] \neq 0$$

证毕.

但し、$|\langle\alpha'|\alpha\rangle|^2 = \exp\left[-|\alpha|^2 - |\alpha'|^2 + \alpha\alpha^{*\prime} + \alpha^*\alpha'\right]$

$= \exp\left[-|\alpha - \alpha'|^2\right]$,

当 $|\alpha - \alpha'| \gg 1$ 時 $\to 0$

(b) 完備性.

$$\int \frac{d^2\alpha}{\pi} |\alpha\rangle\langle\alpha| = \mathbb{1}$$

・直交座標

$\alpha = x + iy$, $d^2\alpha \equiv d(\mathrm{Re}\alpha) \cdot d(\mathrm{Im}\alpha)$

$= dx\,dy$ (x, y: 実数)

・極座標

$\alpha = |\alpha| e^{i\phi} \equiv r e^{i\phi}$, $d^2\alpha = r\,dr\,d\phi$

(r: 正実数, $0 \le \phi \le 2\pi$)

· 证明：由 $|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$,

$$\int \frac{d^2\alpha}{\pi} |\alpha\rangle\langle\alpha| = \int \frac{d^2\alpha}{\pi} \left(e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \right) \left(e^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{(\alpha^*)^m}{\sqrt{m!}} \langle m| \right)$$

$$= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \int \frac{d^2\alpha}{\pi} \; e^{-|\alpha|^2} \; \frac{\alpha^n (\alpha^*)^m}{\sqrt{n!\,m!}} \; |n\rangle\langle m|$$

采用极坐标：
$$\begin{cases} \alpha = r e^{i\phi}, \; \alpha^* = r e^{-i\phi} \\ d^2\alpha = r\,dr\,d\phi \end{cases}$$

$$= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{1}{\pi} \int_0^{\infty} \int_0^{2\pi} r\,dr\,d\phi \; \frac{e^{-r^2} r^{n+m}}{\sqrt{n!\,m!}} e^{i(n-m)\phi} \; |n\rangle\langle m|$$

$$= 2 \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \int_0^{\infty} r\,dr \; \frac{e^{-r^2} r^{n+m}}{\sqrt{n!\,m!}} |n\rangle\langle m| \left[\underbrace{\frac{1}{2\pi} \int_0^{2\pi} d\phi \; e^{i(n-m)\phi}}_{\delta_{nm}} \right]$$

$$= \sum_{n=0}^{\infty} |n\rangle\langle n| = \mathbb{1}$$

$$= \sum_{n=0}^{\infty} \frac{i\omega}{1} \; \widetilde{i^n} \; |n\rangle\langle n|$$

$$= \sum_{n=0}^{\infty} \frac{i\omega}{n!} \; \widetilde{\Gamma(n+1)} \; |n\rangle\langle n|$$

Γ-함수

$$\text{적분 변수} \quad \Gamma(z) \equiv \int_0^\infty dy \; y^{z-1} e^{-y}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \int_0^\infty dy \; y^n e^{-y} \; |n\rangle\langle n|$$

$$\sqrt{\;} \quad r^2 = y$$

$$= 2 \sum_{n=0}^{\infty} \int_0^\infty r\,dr \; e^{-r^2} \frac{r^{2n}}{n!} \; |n\rangle\langle n| \quad .$$

· 基矢间关系

$$|\alpha\rangle = \mathbb{1}|\alpha\rangle = \int \frac{d^2\alpha'}{\pi} |\alpha'\rangle\langle\alpha'|\alpha\rangle$$

$$= \int \frac{d^2\alpha'}{\pi} |\alpha'\rangle \exp\left[-\frac{1}{2}\left(|\alpha|^2 + |\alpha'|^2 - 2\alpha'^*\alpha\right)\right]$$

展开系数 $\neq \delta(\alpha-\alpha')$.

$\Rightarrow$ 完备集中的任意基矢 $|\alpha\rangle$，可以用所有基矢 $|\alpha'\rangle$ 的线性组合表示，即基矢间线性相关！

$\Rightarrow$ 超完备.

2. 态矢和算符的相干态表示式

(a) 态矢：$|\psi\rangle$

$$|\psi\rangle = \mathbb{1}|\psi\rangle = \int \frac{d^2\alpha}{\pi} |\alpha\rangle\langle\alpha|\psi\rangle .$$

$\langle\alpha|\psi\rangle$：态 $|\psi\rangle$ 的相干态表象

$$\langle\alpha|\psi\rangle = \sum_{n=0}^{\infty} \langle\alpha|n\rangle\langle n|\psi\rangle$$

$$= \sum_{n=0}^{\infty} \left(\frac{(\alpha^*)^n}{\sqrt{n!}} e^{-|\alpha|^2/2} \right) \langle n|\psi\rangle$$

$$= \left[\sum_{n=0}^{\infty} \frac{(\alpha^*)^n}{\sqrt{n!}} \langle n|\psi\rangle \right] e^{-|\alpha|^2/2}$$

$$\equiv \psi(\alpha^*)\, e^{-|\alpha|^2/2}$$

$$|\psi\rangle = \int \frac{d^2\alpha}{\pi} |\alpha\rangle\, \psi(\alpha^*)\, e^{-|\alpha|^2/2}$$

Bi: $|\psi\rangle = |n\rangle$

則, $\psi(\alpha^*) = (\alpha^*)^n/\sqrt{n!}$

- $|n\rangle = \int \frac{d^2\alpha}{\pi} (\alpha^*)^n/\sqrt{n!}\ e^{-\frac{|\alpha|^2}{2}} |\alpha\rangle$
- $|0\rangle = \int \frac{d^2\alpha}{\pi}\ e^{-\frac{|\alpha|^2}{2}}\ |\alpha\rangle$

· 內積: $\langle\phi|\psi\rangle$

$$\langle\phi|\psi\rangle = \langle\phi| \left(\int \frac{d^2\alpha}{\pi} |\alpha\rangle\langle\alpha| \right) |\psi\rangle$$

$$= \int \frac{d^2\alpha}{\pi}\ \langle\phi|\alpha\rangle\langle\alpha|\psi\rangle$$

$$= \int \frac{d^2\alpha}{\pi}\ \left[\ \phi^*(\alpha^*)\ e^{-|\alpha|^2/2}\ \right] \times \left[\ \psi(\alpha^*)\ e^{-|\alpha|^2/2}\ \right]$$

$$= \int \frac{d^2\alpha}{\pi}\ \phi^*(\alpha^*)\,\psi(\alpha^*)\ e^{-|\alpha|^2}.$$

(b) 算符的相干态表示

$$\hat{A} = \mathbb{1}\hat{A}\mathbb{1} = \int \frac{d^2\alpha}{\pi} |\alpha\rangle\langle\alpha| \hat{A} \int \frac{d^2\alpha'}{\pi} |\alpha'\rangle\langle\alpha'|$$

$$= \int \frac{d^2\alpha}{\pi} \int \frac{d^2\alpha'}{\pi} \langle\alpha|\hat{A}|\alpha'\rangle |\alpha\rangle\langle\alpha'|$$

$$= \int \frac{d^2\alpha}{\pi} \int \frac{d^2\alpha'}{\pi} A_{\alpha\alpha'} |\alpha\rangle\langle\alpha'|$$

$$A_{\alpha\alpha'} = \langle\alpha|\hat{A}|\alpha'\rangle$$

$$= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \langle\alpha|m\rangle\langle m|\hat{A}|n\rangle\langle n|\alpha'\rangle$$

$$= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \langle\alpha|m\rangle A_{mn} \langle n|\alpha'\rangle$$

$$= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(\alpha^*)^m}{\sqrt{m!}} \frac{(\alpha')^n}{\sqrt{n!}} e^{-|\alpha|^2/2} e^{-|\alpha'|^2/2} A_{mn}$$

$$\equiv A(\alpha^*, \alpha') e^{-|\alpha|^2/2} e^{-|\alpha'|^2/2}$$

$$\Rightarrow \hat{A} = \int \frac{d^2\alpha}{\pi} \int \frac{d^2\alpha'}{\pi} \left[A(\alpha^*, \alpha') e^{-|\alpha|^2/2} e^{-|\alpha'|^2/2} \right] |\alpha\rangle\langle\alpha'|$$

(C). 算符矩阵的迹的相干态表示

(可类比无穷)

$$\mathrm{tr}[\hat{A}] = \sum_{n=0}^{\infty} \langle n|\hat{A}|n\rangle$$

$$= \sum_{n=0}^{\infty} \langle n| \left(\int \frac{d^2\alpha}{\pi} |\alpha\rangle\langle\alpha| \right] \hat{A} |n\rangle$$

$$= \sum_{n=0}^{\infty} \int \frac{d^2\alpha}{\pi} \langle n|\alpha\rangle\langle\alpha|\hat{A}|n\rangle$$

$$= \int \frac{d^2\alpha}{\pi} \sum_{n=0}^{\infty} \langle\alpha|\hat{A}|n\rangle\langle n|\alpha\rangle$$

$$= \int \frac{d^2\alpha}{\pi} \langle\alpha|\hat{A}|\alpha\rangle$$

$$= \int \frac{d^2\alpha}{\pi} A_{\alpha\alpha} .$$

(d) $\hat{A}$ 为正规乘积算符的情况

$\hat{A} = \hat{A}(\hat{a}^\dagger, \hat{a})$，所有 $\hat{a}^\dagger$ 都在所有 $\hat{a}$ 的左边：

（可展成级数）

$(\hat{a}^\dagger)^n (\hat{a})^m$ 等.

- $|\alpha\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger} |0\rangle$

$= e^{-\frac{\alpha^* \alpha}{2}} e^{\alpha \hat{a}^\dagger} |0\rangle$.

- $\frac{\partial}{\partial \alpha} |\alpha\rangle = \left[-\frac{\alpha^*}{2} + \hat{a}^\dagger \right] |\alpha\rangle$.

$\Rightarrow \quad \left(\frac{\partial}{\partial \alpha} + \frac{\alpha^*}{2} \right) |\alpha\rangle = \hat{a}^\dagger |\alpha\rangle$.

共轭式：

$$\langle \alpha | \hat{a} = \left(\frac{\partial}{\partial \alpha^*} + \frac{\alpha}{2} \right) \langle \alpha | .$$

另外，$\langle \alpha | \hat{a}^\dagger = \alpha^* \langle \alpha |$，（$\hat{a}^\dagger |\alpha\rangle = \alpha |\alpha\rangle$）

$$\Rightarrow \quad \langle\alpha|\, \hat{A}_{\text{正規}}(\hat{a}^\dagger, \hat{a})$$

$$= A_{\text{正規}}\left(\alpha^*, \left(\frac{\partial}{\partial\alpha^*} + \frac{\alpha}{2}\right)\right)\langle\alpha|$$

$$\Rightarrow \quad \langle\alpha|\, \hat{A}_{\text{正規}}(\hat{a}^\dagger, \hat{a})\,|\alpha'\rangle$$

$$= A_{\text{正規}}\left(\alpha^*, \left(\frac{\partial}{\partial\alpha^*} + \frac{\alpha}{2}\right)\right)\langle\alpha|\alpha'\rangle$$

$$\hat{A}_{\text{正規}}(\hat{a}^\dagger, \hat{a}) = \int \frac{d^2\alpha}{\pi} \int \frac{d^2\alpha'}{\pi} |\alpha\rangle\langle\alpha|\, \hat{A}_{\text{正規}}(\hat{a}^\dagger, \hat{a})\,|\alpha'\rangle\langle\alpha'|$$

$$= \int \frac{d^2\alpha'}{\pi} \left[\int \frac{d^2\alpha'}{\pi} |\alpha\rangle \left[A_{\text{正規}}\left(\alpha^*, \left(\frac{\partial}{\partial\alpha^*} + \frac{\alpha}{2}\right)\right)\langle\alpha|\alpha'\rangle\right]\langle\alpha'|\right]$$

$$= \int \frac{d^2\alpha'}{\pi} |\alpha\rangle\, A_{\text{正規}}\left(\alpha^*, \left(\frac{\partial}{\partial\alpha^*} + \frac{\alpha}{2}\right)\right)\langle\alpha|$$

3. 在相干态表象中求解一维线性谐振子

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 \hat{x}^2}{2} = \left(\hat{a}^+\hat{a} + \frac{1}{2}\right)\hbar\omega .$$

$$\hat{H}\,|E\rangle = E\,|E\rangle .$$

- $$\langle\alpha|\hat{H}|E\rangle = E\,\langle\alpha|E\rangle .$$

$$\Rightarrow \int \frac{d\alpha'^2}{\pi} \langle\alpha|\hat{H}|\alpha'\rangle\langle\alpha'|E\rangle = E\langle\alpha|E\rangle$$

但，$$\langle\alpha|\hat{H}|\alpha'\rangle = \langle\alpha|\left(\hat{a}^+\hat{a} + \frac{1}{2}\right)|\alpha'\rangle$$

$$\hbar\omega$$

$$= \left[\alpha^*\left(\frac{\partial}{\partial\alpha^*} + \frac{\alpha}{2}\right) + \frac{1}{2}\right]\hbar\omega\,\langle\alpha|\alpha'\rangle$$

$$\Rightarrow \quad \int \frac{d\alpha'^2}{\pi} \left[\alpha^* \left(\frac{\partial}{\partial \alpha^*} + \frac{\alpha}{2} \right) + \frac{1}{2} \right] \hbar\omega$$

$$\langle \alpha | \alpha' \rangle \langle \alpha' | E \rangle = E \langle \alpha | E \rangle .$$

$$\left[\alpha^* \left(\frac{\partial}{\partial \alpha^*} + \frac{\alpha}{2} \right) + \frac{1}{2} \right] \hbar\omega \langle \alpha | E \rangle = E \langle \alpha | E \rangle$$

解：
$$\begin{cases} \langle \alpha | E \rangle = C \, e^{-\frac{|\alpha|^2}{2}} (\alpha^*)^n , \\ E = \left(n + \frac{1}{2} \right) \hbar\omega , \end{cases}$$

C：归一化常数

n：待定实数

令 $\alpha = |\alpha| e^{i\phi}$, $(0 \le \phi \le 2\pi)$

$$\Rightarrow \quad \langle \alpha | E \rangle = C \, e^{-|\alpha|^2/2} \, |\alpha|^n \, e^{-i\phi n}$$

① $\langle\alpha|E\rangle$ 的单值性

$\phi \to \phi + 2\pi$ 时 $\langle\alpha|E\rangle$ 不变.

$(e^{i2\pi n} = 1)$

$\Rightarrow$ n 为包括零的正、负整数.

② $\langle\alpha|E\rangle$ 的有限性

$|\alpha| \to 0$ 时 $\langle\alpha|E\rangle$ 收敛.

$\Rightarrow$ n 非负.

$n = 0, 1, 2, \cdots$

· $\langle E|E\rangle = 1$

$$\int \frac{d^2\alpha}{\pi} \langle E|\alpha\rangle\langle\alpha|E\rangle = 1$$

$$\Rightarrow \quad C = \frac{1}{\sqrt{n!}}$$

· 本征值： $E = E_n = (n+\frac{1}{2})\hbar\omega$

$n = 0, 1, \cdots$

· 本征态： $\langle\alpha|E\rangle = \langle\alpha|E_n\rangle$

$$= \frac{1}{\sqrt{n!}} e^{-|\alpha|^2/2} (\alpha^*)^n$$

※ 相干态表象中展开系数非唯一！

例：求谐振子基态 $|\psi\rangle=|0\rangle$ 的

相干态展开式

① 利用正规展开法

$$|\psi\rangle=\int\frac{d^2\alpha}{\pi}|\alpha\rangle\,\psi(\alpha^*)\,e^{\frac{|\alpha|^2}{2}},$$

其中 $\psi(\alpha^*)=\sum_{n=0}^{\infty}\frac{(\alpha^*)^n}{\sqrt{n!}}\langle n|\psi\rangle$

当 $|\psi\rangle=|0\rangle$ 时，

$$\psi(\alpha^*)=\sum_{n=0}^{\infty}\frac{(\alpha^*)^n}{\sqrt{n!}}\langle n|0\rangle=1$$

$\Rightarrow$

$$|0\rangle = \int \frac{d^2\alpha}{\pi} |\alpha\rangle \underbrace{e^{-\frac{|\alpha|^2}{2}}}$$ 展开系数

右边：　$(\alpha \equiv r e^{i\theta})$

$$\int \frac{d^2\alpha}{\pi} |\alpha\rangle e^{-\frac{|\alpha|^2}{2}} = \frac{1}{\pi} \int_0^\infty r dr \int_0^{2\pi} d\theta$$

$$\times \left(\sum_{n=0}^{\infty} \frac{e^{-\frac{r^2}{2}}}{\sqrt{n!}} r^n e^{in\theta} |n\rangle \right) e^{-\frac{r^2}{2}}$$

$$\underbrace{\qquad\qquad |\alpha\rangle \qquad\qquad}$$

$$= 2 \int_0^\infty r dr \sum_{n=0}^{\infty} r^n e^{-r^2} \frac{1}{2\pi} \underbrace{\int_0^{2\pi} d\theta\, e^{in\theta}}_{\delta_{n0}} |n\rangle$$

$$= 2 \int_0^\infty r dr\, e^{-r^2} |0\rangle = \Gamma(1) |0\rangle$$

$$= |0\rangle .$$

(2) $|0\rangle = \int \frac{d^2\alpha}{\pi} |\alpha\rangle \underbrace{c_1 r}_{\text{展开系数}}$ $(\alpha = re^{i\theta})$

其中 $c_1 \equiv \frac{1}{2\int_0^\infty dr\, r^2 e^{-r^2/2}}$ 为常数

证明：

$$\int \frac{d^2\alpha}{\pi} |\alpha\rangle c_1 r$$

$$= \int \frac{d^2\alpha}{\pi} \left[e^{-r^2/2} \sum_{n=0}^{\infty} \frac{r^n e^{in\theta}}{\sqrt{n!}} |n\rangle \right] c_1 r$$

$$= 2\sum_{n=0}^{\infty} \frac{c_1}{\sqrt{n!}} \int_0^\infty r\,dr\, e^{-r^2/2} r^{n+1} \underbrace{\frac{1}{2\pi}\int_0^{2\pi} d\theta\, e^{in\theta}}_{\delta_{n,0}} |n\rangle$$

$$= 2c_1 \int_0^\infty dr\, r^2 e^{-\frac{r^2}{2}} |0\rangle$$

$$= c_1 / c_1 |0\rangle = |0\rangle \quad \text{ok!}$$

(3) $|0\rangle = \int \frac{d^2\alpha}{\pi} |\alpha\rangle \underbrace{c_n r^n}_{\text{展开系数}}$

（$n > 1$ 的正整数）

· 正规展开的优点 （全纯表象）

$$|\psi\rangle = \int \frac{d^2\alpha}{\pi} |\alpha\rangle \psi(\alpha^*) e^{-|\alpha|^2/2}$$

$|\psi\rangle$ 的展开系数

证：

$$\langle \alpha' | \psi \rangle = \int \frac{d^2\alpha}{\pi} \langle \alpha' | \alpha \rangle \psi(\alpha^*) e^{-|\alpha|^2/2}$$

$$= \psi(\alpha'^*) e^{-|\alpha'|^2/2}$$

展开系数.

即展开系数等于态的相干态表象.

（类似于正交完备集情况）

· 上式证明利用关系式：

$$(\alpha^*)^n = \int \frac{d^2\alpha'}{\pi} \exp\left[-|\alpha'|^2 + \alpha^* \alpha'\right] (\alpha'^*)^n$$

(1) $|0\rangle = \int \frac{d^2\alpha}{\pi} |\alpha\rangle \, e^{-|\alpha|^2/2}$

$\langle x'|0\rangle = \int \frac{d^2\alpha}{\pi} \langle x'|\alpha\rangle \, e^{-|\alpha|^2/2}$

$= e^{-|\alpha'|^2/2}$

(2) $|0\rangle = \int \frac{d^2\alpha}{\pi} \boxed{|\alpha\rangle \, c.r}$

$\langle\alpha'|0\rangle = \int \frac{d^2\alpha}{\pi} \langle\alpha'|\alpha\rangle \, c.r$

$= e^{-|\alpha'|^2/2}$

- 展开系数不等于态的相干态表象！

§1.7 相干压缩态 (coherent squeezed state)

- $[\hat{x}, \hat{p}] = i\hbar$ (基本假设) $\hat{H} = (\hat{a}^\dagger \hat{a} + \frac{1}{2})\hbar\omega$

- 引入 $\hat{a}, \hat{a}^\dagger$, $([\hat{a}, \hat{a}^\dagger] = 1)$

$$\begin{cases} \hat{x} \equiv \sqrt{\frac{\hbar}{2m\omega}}\,(\hat{a}^\dagger + \hat{a}) \\ \hat{p} \equiv i\sqrt{\frac{m\hbar\omega}{2}}\,(\hat{a}^\dagger - \hat{a}) \end{cases}$$

- 相干态 $|\alpha\rangle$: 湮灭算符 $\hat{a}$ 的本征态.

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle \qquad \text{(正则相干态)}$$

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle$$

其中, $\hat{D}(\alpha) \equiv e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$, (位移算符)

$|0\rangle$: 粒子真空, $\hat{a}|0\rangle = 0$.

· $|\alpha\rangle$ 是具有最小不确定性的量子态.

Heisenberg 不确定关系：

$$[\hat{x}, \hat{p}] = i\hbar$$

$$\begin{cases} \overline{(\Delta\hat{x})^2} \equiv \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 \\ \overline{(\Delta\hat{p})^2} \equiv \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2 \end{cases}$$

则 $\underline{\overline{(\Delta\hat{x})^2} \cdot \overline{(\Delta\hat{p})^2} = \frac{\hbar^2}{4}}$

若 $\underline{\overline{(\Delta\hat{x})^2} \cdot \overline{(\Delta\hat{p})^2} = \frac{\hbar^2}{4}}$ （具有最小不确定性的量子态.）

已经验证：

$$\Rightarrow \begin{cases} \overline{(\Delta\hat{x})^2}_{|\alpha\rangle} = \frac{\hbar}{2m\omega} \\ \overline{(\Delta\hat{p})^2}_{|\alpha\rangle} = \frac{m\omega\hbar}{2} \\ \overline{(\Delta\hat{x})^2}_{|\alpha\rangle} \cdot \overline{(\Delta\hat{p})^2}_{|\alpha\rangle} = \frac{\hbar^2}{4}. \end{cases}$$

- 相干压缩态

$$\overline{(\Delta\hat{X})_s^2} < \frac{\hbar}{2m\omega} \quad \left(\text{or} \quad \overline{(\Delta\hat{p})_s^2} < \frac{m\hbar\omega}{2} \right)$$

且 $$\overline{(\Delta\hat{X})_s^2} \cdot \overline{(\Delta\hat{p})_s^2} = \frac{\hbar^2}{4} !$$

1) 压缩算符及压缩变换

(a) 压缩算符 $\hat{S}(\xi)$:

$$\hat{S}(\xi) \equiv \exp\left[\frac{1}{2} \left(\xi^* \hat{a}^2 - \xi (\hat{a}^+)^2 \right) \right]$$

其中 ξ 为复数 : $\xi \equiv r \exp[i\theta]$

$0 < r = |\xi| < \infty \quad , \quad 0 \le \theta \le 2\pi$

压缩因子　　压缩角

- 压缩算符是幺正算符

$$\hat{S}^{\dagger}(\xi) = \hat{S}(-\xi) = \hat{S}^{-1}(\xi)$$

(b) 压缩变换

引入新算符

$$\hat{b} \equiv \hat{S}(\xi)\,\hat{a}\,\hat{S}^{+}(\xi)$$
$$= \hat{a}\,\mathrm{ch}r + \hat{a}^{\dagger} e^{i\theta}\,\mathrm{sh}r ,$$

$$\hat{b}^{\dagger} \equiv \hat{S}(\xi)\,\hat{a}^{\dagger}\,\hat{S}^{+}(\xi)$$
$$= \hat{a}^{\dagger}\,\mathrm{ch}r + \hat{a}\,e^{-i\theta}\,\mathrm{sh}r .$$

利用3算符公式：

$$e^{\hat{A}}\hat{B}\,e^{-\hat{A}} = \hat{B} + [\hat{A},\hat{B}] + \frac{1}{2}\left[\hat{A},[\hat{A},\hat{B}]\right] + \cdots .$$

- 对易关系：

$$[\hat{b}, \hat{b}^\dagger] = \hat{S}[\hat{a}, \hat{a}^\dagger]\hat{S}^\dagger = \hat{S}\hat{S}^\dagger = \mathbb{1}$$

- 新的粒子数算符：

$$\hat{N}_b \equiv \hat{b}^\dagger\hat{b} = \hat{S}\hat{a}^\dagger\hat{a}\hat{S}^\dagger = \hat{S}\hat{N}\hat{S}^\dagger$$

- 本征矢方程：

$$\hat{N}_b|n_b\rangle = n_b|n_b\rangle$$

- $\begin{cases} \text{本征态：} |n_b\rangle = \hat{S}|n\rangle \qquad (\hat{N}|n\rangle = n|n\rangle) \\ \text{本征值：} n_b = n = 0, 1, 2, \cdots \end{cases}$

证明：

$$\hat{N}_b|n_b\rangle = \hat{b}^\dagger\hat{b}\hat{S}|n\rangle$$

$$= \hat{S}\hat{S}^\dagger\hat{b}^\dagger\hat{b}\hat{S}|n\rangle$$

$$= \hat{S}\hat{a}^\dagger\hat{a}|n\rangle = \hat{S}\hat{N}|n\rangle$$

$$= n\hat{S}|n\rangle = n|n_b\rangle .$$

$$\begin{cases} \hat{b}^\dagger|n_b\rangle = \sqrt{n_b+1}\,|n_b+1\rangle \\ \hat{b}|n_b\rangle = \sqrt{n_b}\,|n_b-1\rangle . \quad (n_b\neq 0) \end{cases}$$

$$\hat{b}|n_b=0\rangle = 0 \quad (\text{零矢量}) .$$

定义 $|n_b=0\rangle = \hat{S}|n=0\rangle \equiv |0\rangle_b$ (新粒子真空态)

则有, $\hat{b}|0\rangle_b = 0$

- 引入新的态：

$$|\alpha,\xi\rangle_s \equiv \hat{D}(\alpha)\hat{S}(\xi)|0\rangle$$
$$= \hat{D}(\alpha)|0\rangle_b$$

其中，$\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$ 为以前定义的位移算符。

$$\Rightarrow \quad \alpha \hat{a}^\dagger - \alpha^* \hat{a}$$

$$= \alpha \left[\text{chr}\, \hat{b}^\dagger - \text{shr}\, e^{-i\theta} \hat{b} \right]$$

$$- \alpha^* \left[\text{chr}\, \hat{b} - \text{shr}\, e^{i\theta} \hat{b}^\dagger \right]$$

$$= \hat{b}^\dagger \underbrace{\left[\alpha\, \text{chr} + \alpha^* \text{shr}\, e^{i\theta} \right]}_{\equiv \beta}$$

$$- \hat{b} \underbrace{\left[\alpha^* \text{chr} + \alpha\, \text{shr}\, e^{i\theta} \right]}_{\beta^*}$$

$$= b^\dagger \beta - \hat{b} \beta^*$$

$$\Rightarrow \quad \alpha \hat{a}^\dagger - \alpha^* \hat{a}$$

$$\Rightarrow \quad \underline{D(\alpha)} = e^{\beta \hat{b}^\dagger - \beta^* \hat{b}} \equiv \underline{\hat{D}_b(\beta)}.$$

$$\Rightarrow \quad \underline{|\alpha, \xi\rangle_s = \hat{D}_b(\beta) |0\rangle_b}$$

- $\hat{D}_b(\beta)$：新的平移算符。（幺正算符）

  - $\hat{D}_b(\beta)\,\hat{b}\,\hat{D}_b^\dagger(\beta) = \hat{b} - \beta$，（$\hat{D}_b^\dagger \hat{b} \hat{D}_b = \hat{b} + \beta$）

  - $\hat{D}_b(\beta)\,\hat{b}^\dagger\,\hat{D}_b^\dagger(\beta) = \hat{b}^\dagger - \beta^*$，

  ⋮

  .

※ $|\alpha,\beta\rangle_s \equiv \hat{D}_b(\beta)|0\rangle_b$ 是 $\hat{b}$ 的本征值为 β 的本征态！

$$\hat{b}\,|\alpha,\beta\rangle_s = \beta\,|\alpha,\beta\rangle_s$$

or

$$\hat{b}\,\hat{D}_b(\beta)|0\rangle_b = \beta\,\hat{D}_b(\beta)|0\rangle_b$$

验证：

$$\hat{b}\,\hat{D}_b(\beta)|0\rangle_b = \hat{D}_b(\beta)\,\hat{D}_b^\dagger(\beta)\,\hat{b}\,\hat{D}_b(\beta)|0\rangle_b$$

$$= \hat{D}_b(\beta)\left(\hat{b}^\dagger + \beta\right)|0\rangle_b$$

$$= \beta\,\hat{D}_b(\beta)|0\rangle_b \quad .$$

$$\hat{B}|\beta_2\rangle = \beta\,|\beta_2\rangle$$

$$\Rightarrow \quad \hat{B}(a_1,|a_2|=|a_1,\beta_2\rangle \equiv |\beta_2\rangle$$

$$\langle \beta_1|\beta_2\rangle = \langle 0|\hat{B}^{\dagger}(\beta)\cdot\hat{B}(\beta)|0\rangle$$

$$= \langle 0|\,0\rangle = 1 \qquad \text{단위정규 !}$$

$$\therefore \text{단위정규 대각화}$$

3. 算符的平均值与不确定关系

(a) $\langle\beta|\hat{a}|\beta\rangle_b$

$= {}_b\langle\beta|\left[\mathrm{chr}\,\hat{b} - \mathrm{shr}\,e^{i\theta}\,\hat{b}^+\right]|\beta\rangle_b$

$= \mathrm{chr}\,\beta - \mathrm{shr}\,e^{i\theta}\,\beta^*$

$= \mathrm{chr}\left(\alpha\,\mathrm{chr} + \alpha^*\,\mathrm{shr}\,e^{i\theta}\right)$

$- \mathrm{shr}\,e^{i\theta}\left(\alpha^*\,\mathrm{chr} + \alpha\,\mathrm{shr}\,e^{-i\theta}\right)$

$= \alpha\left[\mathrm{chr}^2 - \mathrm{shr}^2\right]$

$= \alpha$

(b) $\langle\beta|\hat{a}^{\dagger}|\beta\rangle_b = \alpha^*$.

(c) $\langle\beta|\hat{a}^{\dagger}\hat{a}|\beta\rangle_b = sh^2 r + |\alpha|^2$.

(d) $\langle\beta|\hat{a}^2|\beta\rangle_b = \alpha^2 - chr\, e^{i\theta} shr$.

· 关于(d)式的证明

(利用 $|\beta\rangle_b = \hat{D}(\alpha)\hat{S}(\xi)|0\rangle$.)

$\langle\beta|\hat{a}^2|\beta\rangle_b =$

$\langle 0|\hat{S}^{\dagger}(\xi)\hat{D}^{\dagger}(\alpha)\hat{a}^2\hat{D}(\alpha)\hat{S}(\xi)|0\rangle$

$= \langle 0|\hat{S}^{\dagger}(\xi)\hat{D}^{\dagger}(\alpha)\hat{a}\hat{D}(\alpha)\hat{D}^{\dagger}(\alpha)\hat{a}\hat{D}(\alpha)\hat{S}(\xi)|0\rangle$

(利用 $\hat{D}^{\dagger}(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha$)

$= \langle 0|\hat{S}^{\dagger}(\xi)[\hat{a}+\alpha][\hat{a}+\alpha]\hat{S}(\xi)|0\rangle$

$$= \langle 0| \hat{S}^{\dagger}(\xi) \left[\hat{a} + \alpha\right]\left[\hat{a}^{\dagger} + \alpha\right] \hat{S}(\xi) |0\rangle$$

$$= \langle 0| \hat{S}^{\dagger}(\xi) \left(\hat{a}^2 + 2\alpha\hat{a} + \alpha^2\right) \hat{S}(\xi) |0\rangle$$

- 其中为-项: $\langle 0| \hat{S}^{\dagger}(\xi)\, \hat{a}^2\, \hat{S}(\xi) |0\rangle$

$$= \langle 0| \hat{S}^{\dagger}(\xi)\, \hat{a}\, \hat{S}(\xi)\, \hat{S}^{\dagger}(\xi)\, \hat{a}\, \hat{S}(\xi) |0\rangle$$

$$= \langle 0| \left[\hat{a}\cosh r - \hat{a}^{\dagger} e^{i\theta}\sinh r\right]\left[\hat{a}\cosh r - \hat{a}^{\dagger} e^{i\theta}\sinh r\right] |0\rangle$$

($\hat{a}$ 作用于 $\langle 0|$ 为 0；$\hat{a}^{\dagger}$ 作用于 $|0\rangle$ 为 0)

$$= \langle 0| \left(\hat{a}\cosh r\right)\left(-\hat{a}^{\dagger} e^{i\theta}\sinh r\right) |0\rangle$$

$$= -\sinh r \cosh r\, e^{i\theta} \langle 0|\, \hat{a}\, \hat{a}^{\dagger} |0\rangle$$

$$= -\sinh r \cosh r\, e^{i\theta} \langle 0| \left(1 + \hat{a}^{\dagger}\hat{a}\right) |0\rangle$$

$$= -\sinh r \cosh r\, e^{i\theta} \,.$$

· 方法2： $\langle 0|\hat{S}^{+}(\xi)\, 2\alpha\, \hat{a}\, \hat{S}(\xi)|0\rangle$

$= 2\alpha \langle 0|\hat{S}^{+}(\xi)\, \hat{a}\, \hat{S}(\xi)|0\rangle$

$= 2\alpha \langle 0|\left[\hat{a}\cosh r - \hat{a}^{+} e^{i\theta}\sinh r\right]|0\rangle = 0$

· 方法3： $\langle 0|\tilde{S}^{-1}(\xi)\, \alpha^{2}\, \hat{S}(\xi)|0\rangle = \alpha^{2}$

⇒ 三项合起来得.

$\langle \xi_{\alpha}\xi | \hat{a}^{2}|\alpha\cdot\xi\rangle_{S} = \alpha^{2} - \sinh r\cosh r\, e^{i\theta}$

证毕！

而言.

$$\overline{(\Delta\hat{x})_s^2} = \frac{\hbar}{2m\omega}\left[\exp[-2r]\cos^2\frac{\theta}{2} + \exp(2r)\sin^2\frac{\theta}{2}\right],$$

$$\overline{(\Delta\hat{p})_s^2} = \frac{m\hbar\omega}{2}\left[\exp[-2r]\sin^2\frac{\theta}{2} + \exp(2r)\cos^2\frac{\theta}{2}\right].$$

- 相干压缩态

$\overline{(\Delta\hat{x})_s^2} < \frac{\hbar}{2m\omega}$. for $\cos\theta > thr$

$\overline{(\Delta\hat{p})_s^2} < \frac{m\hbar\omega}{2}$. for $\cos\theta < -thr$

(a) $\theta = 0$ 情况

$$\overline{(\Delta\hat{x})_s^2} = \frac{\hbar}{2m\omega}\exp[-2r] \quad \left(< \frac{\hbar}{2m\omega}\right)$$

$$\overline{(\Delta\hat{p})_s^2} = \frac{m\hbar\omega}{2}\exp[2r] \quad \left(> \frac{m\hbar\omega}{2}\right)$$

且 $\overline{(\Delta\hat{x})_s^2}\cdot\overline{(\Delta\hat{p})_s^2} = \frac{\hbar^2}{4}$!

(b) $\theta \approx \pi$ 情况

$$\overline{(\Delta\hat{X})_s^2} \approx \frac{\hbar}{2m\omega} \exp(2r) \quad \left(> \frac{\hbar}{2m\omega}\right)$$

$$\overline{(\Delta\hat{p})_s^2} \approx \frac{m\hbar\omega}{2} \exp(-2r) \quad \left(< \frac{m\hbar\omega}{2}\right)$$

不是 $\overline{(\Delta\hat{X})_s^2} \cdot \overline{(\Delta\hat{p})_s^2} = \frac{\hbar^2}{4}$!

4. $|\alpha,\xi\rangle_s$ 的展开式

$$|\alpha\,\xi\rangle_s = \mathbb{1}\;|\alpha\,\xi\rangle_s = \sum_{n=0}^{\infty} |n\rangle\langle n|\,\alpha\,\xi\rangle_s \,.$$

$$\langle n|\,\alpha\,\xi\rangle_s = \frac{1}{\sqrt{\cosh r}} \exp\left[-\frac{|\alpha|^2}{2} - \frac{\alpha^{2*}}{2} e^{i\theta}\tanh r\right]$$

$$\times \frac{\left[\frac{e^{i\theta}}{2}\tanh r\right]^{n/2}}{\sqrt{n!}} H_n\left[\beta\left(e^{i\theta}\sinh 2r\right)^{-\frac{1}{2}}\right]$$

其中 $H_n(x)$ 为厄米多项式.

- Optical communication
- photon detection techniques
- gravitational wave detection
- ⋮

应用

- 电磁场相干态的制备 (1985, Bell Labs)